Aayush Iyengar

Milwaukee, WI | 262-613-3002 | aayushiyengar2021@gmail.com | linkedin.com/in/aayush-iyengar/

EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 - Present**
M.S in Aerospace Engineering, Concentration in Structural Dynamics and Astrodynamics ***GPA: 4.00 / 4.00***

Purdue University – West Lafayette, IN **August 2022 - May 2026**
B.S in Aeronautical and Astronautical Engineering, Concentration in Astrodynamics & Control; Minor in AI + ML ***GPA: 3.86 / 4.00***

PROFESSIONAL EXPERIENCE

Rocket Lab – *Spacecraft Mechanical Engineer* **June 2026 – Present**

- Designing and building the next generation space vehicles for defense and science missions.

SpaceX – *Starship Stage 0 Propellent Systems Intern* **May 2025 – August 2025**

- Owned the rebuild, robustification, and activation of the static fire stand water farm system, outlining a GN2 bottle blowdown test to proof all newly designed and installed fluids hardware and instrumentation to prove system readiness.
- Engineering and installed the Orbital Pad 2 flame diverter structural shielding, performing static structural and thermal analysis of v3 Booster launch pressure and thermal loads to minimize overpressure effects on the deflector fluid system.
- Ideated and designed 19 LOX/LN2/LCH4 commodity offload stations to deliver cryogenic propellent into the storage tanks across the pad, decreasing offload time by 15% throughout the pad for the Mars Launch Cadence.
- Led the removal of a CO2 vaporizer and sized the appropriate relief valve for the new GN2 press system CONOPS.

Rocket Lab – *Spacecraft Mechanical Engineering Intern* **January 2025 – April 2025**

- Conceptualized, built, and tested a dynamic gravity offloading system to test the deployment of a 15m linear solar array, analyzing frame stability, the static loads within the suspension assembly, and system energy losses from bearings.
- Optimized mass, volume, and energy constraints to determine the ideal layout of Si & GaAs cells for a 5-kW solar array.
- Qualified and redesigned the linear solar array deployment mechanism to minimize frictional losses between linkages.

SpaceX – *Starship Launch Engineering Intern* **May 2024 – August 2024**

- Redesigned the GCH4 pump supply and venting system to facilitate the demolition of a vertical fuel tank and optimize the current gaseous methane supply system while minimizing boil-off losses by moving from 4” to 6” pipe.
- Analyzed the pipe stresses from the discharge side of the numerous LCH4 pumps into the LCH4 Kennedy heat exchanger skid using CAESAR II to model the optimal pipe routing, expansion joint, and pipe support layout for the system.
- Designed a cover for the flame deflector, lowering technician install time by over 5 minutes per use.

LEADERSHIP EXPERIENCE

Automation and Optimization Lab – *UAV Autonomous Charging Mechanisms Lead* **December 2023 – Present**

- Prototyped and 3D-printed a battery swap mechanism, to enable a mid-air autonomous battery exchange system.
- Developing an in-air charging platform with OpenCV to align two drones for autonomous docking and charging; published the results of the in-air charging system to the AIAA SciTech forum.
- Automated the design of a Butterworth filter to eliminate over 99.4% of drone positioning noise during flight tests.
- Constructed a real-time centroid tracking algorithm for large drone positioning in relation to the small drone.

Energy and Transport Sciences Lab – *Energy Storage Analytics Researcher* **August 2023 – January 2024**

- Created a Python ML framework and a Python GUI to visually demonstrate that eVTOL architecture and electrode design have complex implications on thermal safety and degradation statistics for Li-Ion batteries in eVTOLs.
- Utilized connected MATLAB and C++ scripts to collect data from experimental and simulated performances under normal and anomalous Li-ion operating conditions to identify the optimal battery system for eVTOL systems.
- Simulated over 500,000 Li-ion cycles to identify the point where capacity breaks down for different Li-ion systems.

SKILLS/INTERESTS

Programming Languages: Python, MATLAB, C, Java, HTML
Tools: NX, ANSYS Mechanical, Simulink, Nastran, CAESAR II, SolidWorks, Fusion 360, XFLR 5, Teamcenter, Git
Interests: Robotic Systems, Autonomous Docking, Language Processing, Entrepreneurship, Saxophone/Jazz, and Tennis